

functions

1. built in

2. user defined

```
In [2]: sum([1,3])
```

```
Out[2]: 4
```

```
In [3]: print()
```

```
In [4]: x=3 # input
        y=2 #input
        z=x+y
        print(z)
```

```
5
```

```
In [8]: def hello(x,y):
        z=x+y
        return(z)
```

```
In [9]: hello(3,6) # user defined
```

```
Out[9]: 9
```

```
In [10]: hello(10,4)
```

```
Out[10]: 14
```

```
In [12]: def adddd(x,y):
        z=x+y
        return(z)
```

```
In [14]: name="sachi"
        print("hello",name)
```

```
hello sachi
```

```
In [16]: name=str(input("enter your name"))
print("hello",name)
```

```
enter your nameqwertyu
hello qwertyu
```

```
In [53]: def greet(name):
print("hello",name)
```

```
In [54]: greet("dear")
```

```
hello dear
```

```
In [55]: def greet2(name):
z="hello "+name
return(z)
```

```
In [49]: greet2("dear")
```

```
Out[49]: 'hello dear'
```

define a function named happy to check odd and even numbers

```
In [37]: x=50 #input
if x%2==0:
    print(x, "is even")
else:
    print("odd")
```

```
50 is even
```

```
In [59]: def happy(x):
'''happy is a function which is used to check a number odd or even'''
if x%2==0:
    print(x, "is even")
else:
    print("odd")
```

```
In [61]: happy()
```

```
4 is even
```

```
In [41]: happy(6)
```

```
6 is even
```

```
In [65]: #place holders- order matters  
x="appu"  
y=15  
print("my name is",x," and my age is",y)
```

my name is appu and my age is 15

```
In [44]: def hai(x,y):  
        print("my name is",x," and my age is",y)
```

```
In [46]: hai("appu",12) # x,y
```

my name is appu and my age is 12

```
In [47]: hai(12,"appu") # x,y
```

my name is 12 and my age is appu

```
In [ ]:
```

```
In [64]: i=0 #global variable  
sum=0#global variable  
while i<=10:  
    print(i)  
    y=sum+i #local variable  
    i=i+1
```

0
1
2
3
4
5
6
7
8
9
10

```
In [67]: #key word argument  
def hai(x,y):  
    print("my name is",x," and my age is",y)
```

```
In [68]: hai("appu",13)
```

my name is appu and my age is 13

```
In [72]: hai(x="appu",y=13)
```

```
my name is appu and my age is 13
```

```
In [73]: hai(y=24,x="appu")
```

```
my name is appu and my age is 24
```

```
In [86]: #default argument
def hai (x=50):
    print("the number is ", x)
```

```
In [87]: hai()
```

```
the number is 50
```

```
In [90]: #its not default argument
def hai (x):
    print("the number is ", x)
```

```
In [92]: hai() #error cause it is not default
```

```
-----
TypeError                                Traceback (most recent call last)
~\AppData\Local\Temp\ipykernel_12904\2255484400.py in <module>
----> 1 hai() #error cause it is not default

TypeError: hai() missing 1 required positional argument: 'x'
```

```
In [93]: #variable Length
def haiii(*x):
    for i in x:
        print("my name is",i)
```

```
In [94]: haiii("helo","hai")
```

```
my name is helo
my name is hai
```

```
In [100]: x=["appu","ammu"]
for i in x:
    print("hai",i)
```

```
hai appu
hai ammu
```

```
In [103]: /* is used cause multiple values in a variable  
def hello(*x):  
    for i in x:  
        print("my nameis" ,i)
```

```
In [104]: hello("krish","fathi")
```

```
my nameis krish  
my nameis fathi
```

```
In [ ]:
```

```
In [ ]:
```